

SAI KUMAR REDDY ANTHIREDDY

AI/ML Engineer · Generative AI · RAG Pipelines · MLOps

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3 years of experience designing and deploying machine learning and Generative AI solutions for customer analytics and enterprise support automation. Expert in LLM-powered applications, Retrieval-Augmented Generation (RAG) pipelines, and semantic search using LangChain, Hugging Face, PyTorch, and OpenAI APIs. Proven record of shipping production AI on AWS SageMaker, Docker, and Kubernetes with full CI/CD and model monitoring — including a RAG platform that now handles first-line technical support queries.

TECHNICAL SKILLS

Generative AI & LLMs: LangChain, Hugging Face Transformers, RAG, Prompt Engineering, OpenAI API, LlamaIndex, LangGraph, AI Agents

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Keras, CNNs, RNNs, Transformers, Transfer Learning

NLP: BERT, spaCy, Text Classification, Sentiment Analysis, Embedding Models, Semantic Search

Vector Databases: Pinecone, FAISS, ChromaDB

MLOps & Deployment: Docker, Kubernetes, MLflow, FastAPI, REST APIs, GitHub Actions, CI/CD, Model Monitoring

Cloud & Data: AWS SageMaker, Azure, GCP, Pandas, NumPy, ETL Pipelines, Feature Engineering, SQL, Python, Java

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Teledyne Technologies Inc.

Jul 2025 – Present

- Designed an **Intelligent Technical Knowledge & Support Automation Platform** using RAG over enterprise knowledge sources to automate first-line technical support.
- Built an end-to-end RAG pipeline with **LangChain, OpenAI API, FAISS, and Pinecone** to retrieve, rank, and generate context-aware answers from manuals, SOPs, and support docs.
- Fine-tuned Hugging Face Transformer models (BERT) for intent classification and technical query understanding, benchmarked against a keyword-matching baseline.
- Tuned embedding and chunking strategy while managing vector embeddings in Pinecone and ChromaDB, improving retrieval relevance and cutting query latency.
- Implemented agentic workflows with **LangGraph and LlamaIndex** to orchestrate retrieval, reasoning, and tool execution for complex support requests.
- Containerized AI services with Docker and deployed inference on **Kubernetes and AWS SageMaker** with autoscaling and health probes, replacing a manual release process.
- Automated training, deployment, and versioning with **GitHub Actions, CI/CD, and MLflow**, enabling drift-based retraining and continuous monitoring.
- Mentored junior engineers through code reviews while integrating AI capabilities into enterprise applications with product and domain teams.

Junior Machine Learning Engineer | Colt Technology Services

May 2021 – Jun 2023

- Developed a **Customer Retention Intelligence & Churn Prediction Platform** to identify at-risk customers and drive data-driven retention.
- Cleaned and transformed **250K+ customer records** with Pandas and NumPy, improving data quality and automating a previously manual preprocessing step.
- Engineered features and built supervised models with **Scikit-learn and TensorFlow**, evaluated against the incumbent rules-based retention model.
- Improved ETL pipelines integrating usage, billing, and support data from multiple enterprise systems into ML workflows.
- Optimized classifiers using cross-validation, precision/recall/F1, and confusion-matrix analysis to improve prediction reliability.
- Deployed models as **FastAPI REST APIs on AWS**, enabling real-time churn scoring in place of a nightly batch job.

PORTFOLIO PROJECTS

- **Enterprise AI Platform** — github.com/saianthireddy/enterprise-ai-platform — Full-stack AI copilot: intent router over six task handlers, hybrid search, admin analytics, Next.js frontend, Terraform and Kubernetes infrastructure, 50 automated tests.
- **RAG Support Automation** — github.com/saianthireddy/rag-support-automation — Production-style RAG platform: ingestion and chunking, swappable FAISS/Pinecone stores, citation-grounded generation, FastAPI, Docker, and a published retrieval benchmark.
- **Enterprise MLOps Platform** — github.com/saianthireddy/enterprise-mlops-platform — Model lifecycle from commit to monitored endpoint: champion/challenger promotion, versioned registry, PSI/KS drift detection, Kubernetes and SageMaker deployment.
- **webdocs-mcp** — github.com/saianthireddy/webdocs-mcp — Documentation crawler with hybrid BM25 and embedding search over DuckDB, served to LLM agents as an MCP server.
- **Customer Churn Prediction** — github.com/saianthireddy/customer-churn-prediction — End-to-end churn ML: data validation, feature engineering, Logistic Regression / Gradient Boosting pipelines, cross-validated evaluation, and a real-time /predict API with risk bands.

EDUCATION

M.S. in Advanced Data Analytics | University of North Texas, Denton, TX

Aug 2023 – May 2025

B.S. in Computer Science | Osmania University, Hyderabad, India

Jun 2018 – May 2021

CERTIFICATION

AWS Certified Solutions Architect – Associate — [Verify on Credly](#)